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Motion Consistency Evaluation in Unmodified Surveillance Video Data

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1 Problem Definition & Originality

Problem Statement: The video streams produced by surveillance camera systems are continuous, but in real-world applications, there are hidden problems in the video streams caused by environmental or hardware issues that result in low motion, frozen images, or irregular motion patterns. These problems do not necessarily cause traditional object-based alerts but can significantly affect the reliability of the video streams. The project aims to determine the motion consistency in the unaltered surveillance videos by examining the natural motion patterns in the videos.

Originality Claim: *Novel Idea*

- We propose a method for camera health assessment by exploring the temporal patterns of motion in video data. Unlike existing solutions that focus on object and event detection, our approach enables continuous and non-intrusive assessment of real-world surveillance data without the need for object detection or human annotation.

2 Literature Survey & Gap Analysis

- **Existing Solutions:** Recent literature on surveillance video analysis and anomaly detection has leveraged 3D CNNs, Transformers, and diffusion models. These methods typically focus on detecting behavioral anomalies (e.g., violence or theft) in urban and multi-camera environments.
- **Limitations:** However, most systems assume stable video feeds and overlook signal-level impairments. While video quality assessment (VQA) and forgery detection (tampering/duplication) have matured, they are rarely integrated into a continuous camera health assessment framework. Current VAD methods often miss prolonged motion stagnation and lack security measures for heterogeneous video protocols.
- **Research Gap:** Despite these advancements, there is no unified solution for assessing the temporal motion consistency of raw feeds to determine real-time camera health. A critical gap exists for a non-intrusive, consistency-based evaluation that can distinguish between natural low activity and actual stream malfunctions without requiring object-level detection.

3 Proposed Methodology (Technical Knowledge)

Core Concept: The proposed system introduces a **Motion Consistency-Based Surveillance Feed Evaluation Framework**. Instead of concentrating on the detection of unusual behavioral activities, the framework captures the normal process of motion in surveillance videos and verifies whether that motion is consistent. The process involves:

- Spatio-temporal feature extraction from a sequence of video frames.
- Motion representation modeling using frame difference and optical flow.
- Temporal consistency scoring using a sliding time window.

- Deviation detection to locate anomalies such as frozen frames, duplication, protracted motion stagnation, and temporal discontinuities.

The final output is a **Feed Reliability Score**, which reveals the status and health of the surveillance feed.

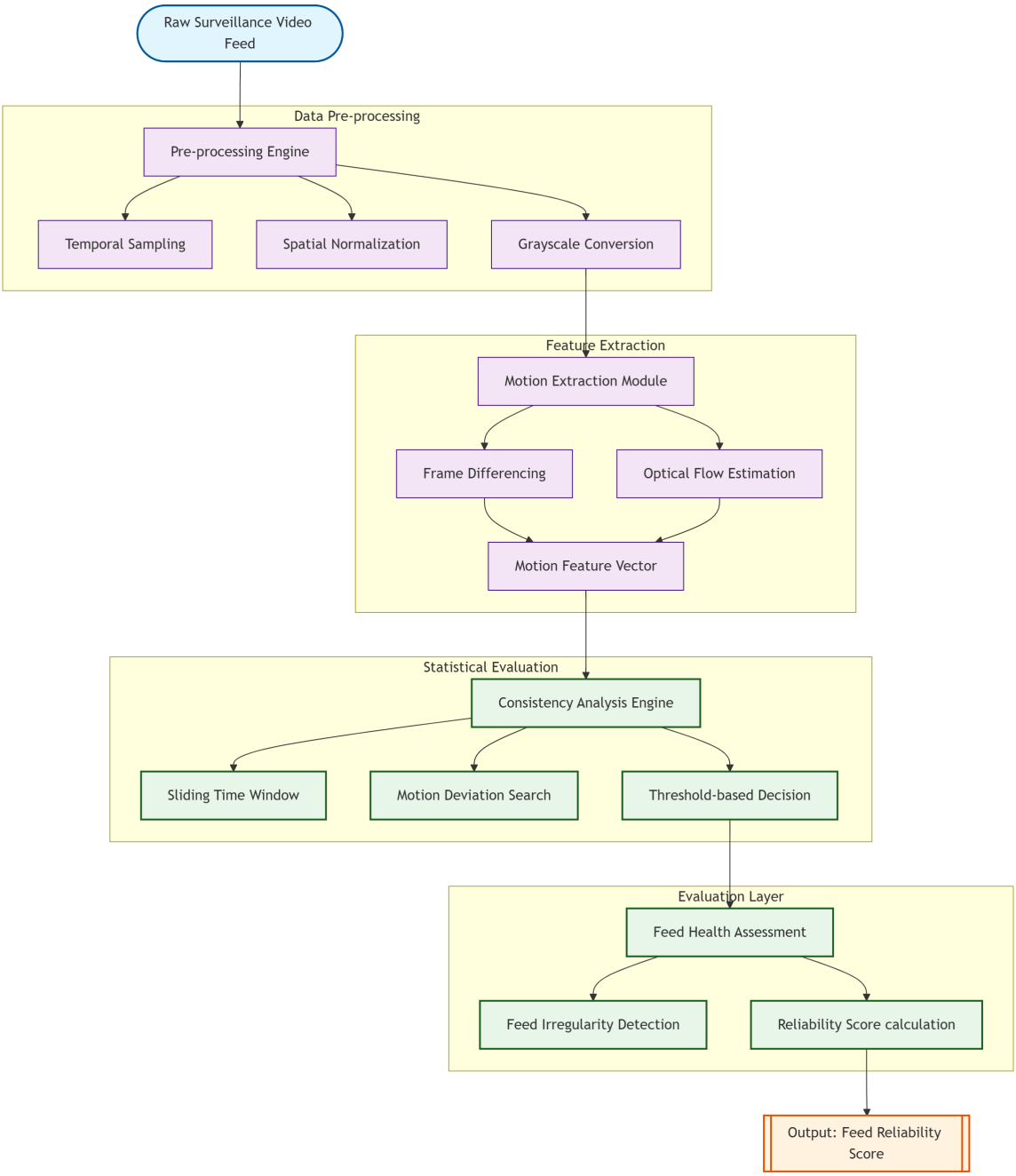

Figure 1: Proposed Methodology Architecture

Mathematical / Logical Basis: The model is based on the measurement of inter-frame motion consistency. Let F_t be the video frame at time t, and M_t represent the motion between F_t and F_{t-1}. Motion is calculated as:

$$M_t = ||F_t - F_{t-1}|| \quad or \quad M_t = \sum ||Flow(F_{t-1}, F_t)|| \tag{1}$$

Step 1: Natural Motion Modeling: Statistical measures are calculated over a sliding window: mean motion (μ), variance (σ^2), and temporal smoothness constraints.

Step 2: Consistency Evaluation: The motion consistency measure C_t is defined as:

$$C_t = \frac{|M_t - \mu_{window}|}{\sigma_{window} + \epsilon} \tag{2}$$

Anomaly detection follows specific rules:

- $M_t \approx 0$ for extended duration $\rightarrow$ Freeze suspicion.
- M_t constant for long window $\rightarrow$ Frame duplication suspicion.
- Sudden abnormal fluctuation $\rightarrow$ Temporal irregularity.

Step 3: Feed Reliability Score: The final score R is calculated by summing consistency deviations:

$$R = 1 - \frac{1}{T} \sum I(C_t > \tau) \tag{3}$$

where τ is a predefined threshold and $I(\cdot)$ is the indicator function.

4 Current Status & Plan

Status: Literature survey completed; Research gap identified; Conceptual model finalized; Architecture designed.
Next Steps: Implementation of the motion extraction module (frame difference/optical flow); Development of a framework for temporal consistency analysis; Design and calculation of feed reliability score; Initial testing and threshold adjustment.

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